MIT 8.012 Problem Set 4 - Problem 3.13 (K&K)

3.13 A ski tow consists of a long belt of rope around two pulleys, one at the bottom of a slope and the other at the top. The pulleys are driven by a husky electric motor so that the rope moves at a steady speed of 1.5 m/s. The pulleys are separated by a distance of 100 m, and the angle of the slope is 20°.

Skiers take hold of the rope and are pulled up to the top, where they release the rope and glide off. If a skier of mass 70 kg takes the tow every 5 s on the average, what is the average force required to pull the rope? Neglect friction between the skis and the snow.

Solution (handwritten) follows on the next pages.

Problem 3

Notation :-

Mass of the person :- M_p

No of people on the rope at time t :- $N(t)$

No of seconds on avg within which one person joins the rope :- k

Length of the incline :- l

velocity at which the rope is moving :- v

Lemma 1 :-

No of people on the rope $N(t) = \frac{t}{vk} \pm \epsilon$

$\forall t \geq \frac{l}{v}$, $|\epsilon| < \frac{1}{2}$

PF :-

Consider $t = \frac{l}{v}$; Nobody has left the rope yet, so the no of people on the rope is :-

$\left\lfloor \frac{l}{vk} \right\rfloor_{Floor}$ Because 1 person joins every k seconds.

Let t_0 be the time when the last person joined in.

Define

We know that $\frac{l}{v} - t_0 \leq k$

Define Two Sequences

$\frac{l}{v}, \frac{l}{v}+K, \frac{l}{v}+2K \ldots\ldots L_n$

$t_0, t_0+K, t_0+2K \ldots\ldots J_n$

L_n represent times when n people leave, J_n represents time when n people join

we Know that $\forall n$

$L_n - J_n \leq K$ and $J_{n+1} - J_n = K$

$J_n < L_n$

So when n people join n people leave before the $(n+1)^{th}$ person joins and vice versa.

So the no of people who join and are on the can atmost be

$$\left\lceil \frac{l}{vK} \right\rceil \text{ and atleast } \left\lfloor \frac{l}{vK} \right\rfloor$$

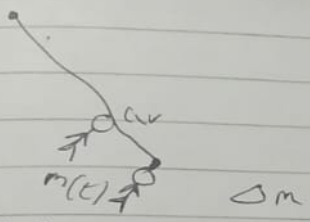
QED.

[Implicit Assumption First person joined at t=s

Analyzing Momentum (Assuming Continuous Mass Transfer)

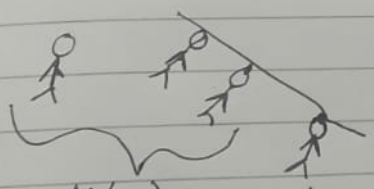
Note: Even Analyzing Momentum in discrete K-length steps would retrieve the same answer, only the notation would differ.

State 1: time t



$\vec{p_{sys}} = m(t)v$

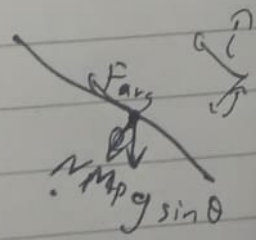
State 2 time $t + \Delta t$



$\vec{p_{sys}}(t+\Delta t) = \vec{v}(m(t) + \Delta m)$

$$\frac{\Delta p}{\Delta t} = v\left(\frac{\Delta m}{\Delta t}\right) = \frac{v m_p}{K} = F_{ext}$$

Force Analysis



$\hat{i}$: $F_{avg} - N M_p g \sin\theta = \frac{v m_p}{K}$

$$F_{avg} = \frac{\ell m_p g \sin\theta}{vK} + \frac{v m_p}{K}$$

$$\approx \frac{100(70)(10)(0.3)}{(5)(1.3)} + 2$$

$$F_{avg} \approx \frac{300(70)}{5(1.5)} + 21$$

$$\approx \frac{\cancel{60}^{\,4}(700)}{(15)} + 21$$

$$\approx 2800 + 21 = 2821\ N$$